

Quiz (Habanero)

- Q1.** Factor the expression $x^2 + 5x + 4x + 20$ by grouping.
(A) $(x + 4)(x + 5)$
(B) $(x + 2)(x + 10)$
(C) $x(x + 5) + 4(x + 5)$
(D) $x(x + 4) + 5(x + 4)$
(E) $x^2 + 9x + 20$
- Q2.** What is the factored form of $3x^2 + 12x + 4x + 16$?
(A) $3(x + 2)(x + 4)$
(B) $3(x + 4)(x + 2)$
(C) $(3x + 4)(x + 4)$
(D) $3x(x + 4) + 4(x + 4)$
(E) $3x(x + 2) + 4(x + 2)$
- Q3.** A city experiences $2x^2 + 5x + 2x + 5$ millimeters of rainfall. Factor the expression.
(A) $2x^2 + 7x + 5$
(B) $2x(x + 2) + 5(x + 1)$
(C) $2x(x + 5) + 1(x + 5)$
(D) $(2x + 5)(x + 1)$
(E) $2(x + 2)(x + 1) + 1$
- Q4.** The temperature in a greenhouse is $x^2 + 2x + 6x + 12$ degrees Celsius. Factor the expression by grouping.
(A) $x^2 + 8x + 12$
(B) $x(x + 6) + 2(x + 6)$
(C) $x(x + 2) + 6(x + 2)$
(D) $(x + 6)(x + 2)$
(E) $(x + 2)(x + 6)$
- Q5.** The length of a shadow of a tree is $x^2 + 3x + 5x + 15$. Factor the expression.
(A) $x^2 + 8x + 15$
(B) $x(x + 5) + 3(x + 5)$
(C) $x(x + 3) + 5(x + 3)$
(D) $(x + 5)(x + 3)$
(E) $(x + 3)(x + 5)$
- Q6.** Factor $2x^2 + 9x + 5x + 10$ by grouping.
(A) $2x^2 + 14x + 10$
(B) $2x(x + 5) + 1(x + 5)$
(C) $2x(x + 9) + 1(x + 10)$
(D) $(2x + 5)(x + 2)$
(E) $2(x + 2)(x + 5)$
- Q7.** A water tank can hold $3x^2 + 2x + 5x + 10$ gallons of water. Factor the expression.
(A) $3x^2 + 7x + 10$
(B) $3x(x + 2) + 5(x + 2)$
(C) $3x(x + 5) + 2(x + 5)$
(D) $3(x + 5)(x + 2)$
(E) $3(x + 2)(x + 5)$
- Q8.** A researcher studies the growth rate of a plant, which is $x^2 + 4x + 3x + 12$. Factor the expression by grouping.
(A) $x^2 + 7x + 12$
(B) $x(x + 3) + 4(x + 3)$
(C) $x(x + 4) + 3(x + 4)$
(D) $(x + 4)(x + 3)$
(E) $(x + 3)(x + 4)$

Open Ended Questions (FRQ)

- Q9.** Factor the expression $2x^2 + 5x + 6x + 15$ by grouping, and verify the factoring by multiplication.
- Q10.** A city experiences a temperature change of $x^2 + 3x + 2x + 6$ degrees Celsius per hour. Factor the expression, and explain the factoring process in 2-3 sentences.

Chef's Tips

- Verify the factoring by multiplication for each question
- Use the given space for calculations and explanations in free-response questions
- Ensure to check the units and context of each problem